JWT Ammonia Recirculator Package: Installation, Operation, and Maintenance Manual (2026 Edition)

Section 1: Introduction and System Overview

1.1 Preface & Manual Scope

This manual provides technical guidance for the safe installation, operation, and maintenance of JWT Recirculator Packages. It is designed to help technicians and engineers achieve safe, reliable, and efficient system performance. The robust construction and quality components are engineered to deliver exceptional performance and a long service life when installed and maintained as directed.

The scope of this document is to serve as the primary technical reference for qualified industrial refrigeration technicians, engineers, and facility managers. Adherence to the procedures outlined herein is essential for ensuring the safety of personnel, protecting the equipment from damage, and achieving the designed operational efficiency. This **2026** Edition has been comprehensively updated to align with ANSI/IIAR 2-2021, IIAR 5-2025 (Startup), IIAR 6-2025 (Inspection, Testing, and Maintenance), ASHRAE 15-2024, and ASME Section VIII Division 1 requirements.

1.2 Principles of Liquid Overfeed (Recirculation)

The JWT Recirculator Package is the core component of a liquid overfeed refrigeration system, a design methodology proven to offer superior performance and safety compared to older direct expansion (DX) systems. Understanding the principles of this system is fundamental to its correct operation.

A liquid overfeed system, also known as a pumped liquid recirculated system, utilizes a mechanical pump to circulate a greater volume of liquid refrigerant through the system's evaporators than is actually evaporated to meet the cooling load. This "overfeed" ensures that the entire internal surface of the evaporator coils remains wetted with liquid refrigerant, maximizing heat transfer efficiency and preventing the performance degradation associated with vapor superheating within the coil itself. A typical design recirculation rate for ammonia systems is 4:1, meaning for every one part of liquid ammonia that is evaporated in the coil, four parts are circulated.

The two-phase (liquid and vapor) mixture returns from the evaporators to the central vessel of the JWT package, which serves as a low-pressure receiver or accumulator. Inside the vessel, the liquid and vapor are separated. The vapor, now dry and at a consistent suction temperature, is drawn off the top of the vessel to the system compressors. This separation process is the primary method of compressor protection, preventing liquid refrigerant "slop-over" or "floodback"—a condition that can cause catastrophic damage to compressor components. The separated liquid refrigerant collects in the bottom of the vessel, providing a ready supply of subcooled liquid with sufficient Net Positive Suction Head (NPSH) for the recirculation pumps. This continuous, robust circulation of liquid refrigerant makes the system less sensitive to fluctuating loads and provides stable, efficient evaporator performance.

1.3 Unit Identification and Design Specifications

Each JWT Recirculator Package is identified by a permanent stainless steel data plate, typically mounted on the vessel shell or the structural steel base. This data plate contains critical information required for safe operation, service, parts ordering, and compliance with regulatory programs such as OSHA's Process Safety Management (PSM) and the EPA's Risk Management Plan (RMP). Before performing any work on the unit, the technician must locate the data plate and verify the equipment's specifications. When contacting JWT for service or parts, always provide the Model Number and Serial Number.

The following table provides a quick-reference summary of key design specifications. For specific values, refer to the unit's data plate and the certified general arrangement drawings provided with the equipment.

Specification	Description
Model Number	The specific model designation of the JWT package.
Serial Number	The unique serial number for tracking and service history.
Vessel MAWP	Maximum Allowable Working Pressure (in psig) of the pressure vessel. **NEVER** exceed this pressure.
MDMT	Minimum Design Metal Temperature (in °F) at the corresponding MAWP.
Refrigerant	Designed for Ammonia (R-717).
Vessel Volume	The internal volume of the accumulator vessel in cubic feet.
National Board No.	The official registration number with the National Board of Boiler and Pressure Vessel Inspectors.
Pump Model	The manufacturer and model number of the installed refrigerant pumps.
Control Voltage	The required voltage for the control panel and its components (e.g., 120V AC).
Pump Motor Specs	The voltage, full-load amperage (FLA), and horsepower of the pump motors.

1.4 Glossary of Terms

- **Accumulator:** A pressure vessel in the low-pressure side of a refrigeration system designed to separate liquid and vapor refrigerant, ensuring only vapor returns to the compressor. Also known as a Low-Pressure Receiver or Surge Drum.
- **ASME:** The American Society of Mechanical Engineers. The governing body for the Boiler and Pressure Vessel Code.
- **ASHRAE:** The American Society of Heating, Refrigerating and Air-Conditioning Engineers.
- **IIAR:** The International Institute of All-Natural Refrigeration. The leading standards-writing organization for the ammonia refrigeration industry.
- **Liquid Overfeed:** A system where the mass flow rate of liquid refrigerant circulated through the evaporator is greater than the rate of evaporation.
- **MAWP (Maximum Allowable Working Pressure):** The maximum gauge pressure permissible at the top of a vessel in its normal operating position at the designated coincident temperature.
- **MDMT (Minimum Design Metal Temperature):** The lowest temperature at which a pressure vessel can be safely operated at its MAWP.
- **NPSH (Net Positive Suction Head):** The pressure at the suction of a pump, over and above the vapor pressure of the liquid, required to prevent cavitation.
- **PID Loop (Proportional-Integral-Derivative):** A control loop mechanism employing feedback that is widely used in industrial control systems.
- **PSM (Process Safety Management):** An OSHA regulation (29 CFR 1910.119) aimed at preventing the catastrophic release of highly hazardous chemicals, including ammonia.
- **RAGAGEP:** Recognized And Generally Accepted Good Engineering Practices. A foundational concept in PSM, often fulfilled by adherence to standards from organizations like IIAR and ASHRAE.
- **Recirculation Rate:** The ratio of the mass of liquid refrigerant circulated to the mass of refrigerant vaporized in the evaporator (e.g., 4:1).
- **RMP (Risk Management Plan):** An EPA regulation (40 CFR Part 68) that requires facilities with more than a threshold quantity of a regulated substance to develop a program to prevent accidental releases.
- **VFD (Variable Frequency Drive):** An electronic device that controls the speed of an AC electric motor by varying the frequency and voltage of the power supplied to it.

Section 2: Safety and Compliance

2.1 Safety Precaution Definitions

The following signal words are used throughout this manual to designate the degree or level of hazard. All personnel working on or near this equipment must understand their meaning and follow all associated instructions.

- **DANGER:** Indicates an imminently hazardous situation which, if not avoided, **will** result in death or serious injury.
- **WARNING:** Indicates a potentially hazardous situation which, if not avoided, **could** result in death or serious injury.
- **CAUTION:** Indicates a potentially hazardous situation which, if not avoided, **may** result in minor or moderate injury or equipment damage.
- **NOTE:** Indicates an operating procedure, practice, or condition that is essential to highlight for proper equipment operation or to prevent damage.

2.2 General Ammonia (R-717) Safety

DANGER: AMMONIA IS A HAZARDOUS AND TOXIC CHEMICAL. EXPOSURE TO HIGH CONCENTRATIONS CAN CAUSE SEVERE INJURY OR DEATH. ONLY TRAINED AND QUALIFIED REFRIGERATION TECHNICIANS SHOULD INSTALL, OPERATE, OR SERVICE THIS EQUIPMENT.

- **Health Hazards:** Ammonia is classified by ASHRAE Standard 34 as a B2L refrigerant, signifying higher toxicity and mild flammability under specific conditions. Inhalation of ammonia vapor can cause severe irritation and damage to the respiratory tract. OSHA's permissible exposure limit (PEL-TWA) for ammonia is 50 ppm for an 8-hour workday. NIOSH and ACGIH recommend a TWA of 25 ppm and a STEL of 35 ppm. The IDLH (Immediately Dangerous to Life or Health) level is 300 ppm. Contact with liquid ammonia will cause severe cryogenic burns and permanent tissue damage.
- **Personal Protective Equipment (PPE):** Appropriate PPE is mandatory when performing any service that creates a potential for ammonia release. This includes, at a minimum, a full-face shield over chemical splash goggles, and ammonia-rated gauntlet-style gloves. Approved respiratory protection (e.g., gas masks with ammonia cartridges) must be readily available and personnel must be trained in their use.

- **Flammability:** While difficult to ignite, ammonia vapor is flammable in concentrations between 16% and 25% by volume in air. All sources of ignition must be eliminated in areas where a significant release could occur.

2.3 Equipment-Specific Safety Hazards

WARNING: HIGH-PRESSURE REFRIGERANT. THIS SYSTEM CONTAINS AMMONIA UNDER HIGH PRESSURE. AN UNCONTROLLED RELEASE CAN CAUSE SERIOUS INJURY OR DEATH. FOLLOW ALL LOTO AND LINE BREAKING PROCEDURES.

- **Hydrostatic Expansion:** Liquid ammonia expands significantly with an increase in temperature. Trapping liquid refrigerant between two closed valves (e.g., on an isolated pump or filter) can generate extreme hydrostatic pressure, leading to the catastrophic rupture of piping, valve bodies, or equipment casings. Always ensure a path to a relief device or expansion volume when isolating components.
- **Electrical Shock Hazard:** This package contains high-voltage electrical components. All electrical work must be performed by qualified electricians. Before servicing, de-energize and lock out/tag out (LOTO) all associated power sources at the main disconnect in accordance with facility safety procedures and OSHA regulations.
- **Rotating Equipment Hazard:** Pumps contain rotating shafts and couplings. Ensure all guards are in place before operation. Before performing maintenance, de-energize and LOTO the pump motor. Keep hands, clothing, and tools clear of all moving parts.
- **Lifting Hazard:** The recirculator package is heavy and may have an offset center of gravity. Lifting must only be performed by qualified riggers using the specified lifting lugs and procedures outlined in Section 3. An unstable lift can result in the unit tipping or falling, which could cause death, serious injury, and equipment damage.

2.4 Regulatory Compliance and 2026 Standards

The JWT Recirculator Package is designed and fabricated to be a fully compliant component within a larger industrial refrigeration system. Its construction and design adhere to Recognized and Generally Accepted Good Engineering Practices (RAGAGEP), making it suitable for facilities governed by OSHA's PSM and EPA's RMP regulations. The procedural guidance within this IOM is intended to serve as a foundational document for the end-user to develop their site-specific Standard Operating Procedures (SOPs) as required by these regulations and industry standards.

The design and fabrication of this unit comply with the latest editions of the following key codes and standards:

- **ANSI/IIAR 2-2021:** *Standard for Design of Safe Closed-Circuit Ammonia Refrigeration Systems*. This is the primary design standard for the ammonia refrigeration industry, covering all aspects of system safety.
- **ANSI/IIAR 9-2020 (Addendum A-2024):** *Standard for Minimum System Safety Requirements for Existing Closed-Circuit Ammonia Refrigeration Systems*. This standard is critical when integrating the JWT package into existing facilities, ensuring the entire system meets a consistent safety baseline.
- **ASME Boiler and Pressure Vessel Code, Section VIII, Division 1:** This code governs the design, materials, fabrication, and testing of the pressure vessel to ensure its integrity under pressure.
- **ASHRAE 15-2024:** *Safety Standard for Refrigeration Systems*. This standard provides overarching safety requirements for refrigeration systems.
- **ANSI/ASME A13.1:** *Scheme for the Identification of Piping Systems*. This standard provides the basis for proper pipe labeling, which is a critical safety feature in any ammonia engine room.

Section 3: Shipping, Handling, and Installation

3.1 Initial Receipt and Inspection

Upon delivery of the equipment, immediately perform a thorough inspection of all crates, boxes, and exposed component surfaces for any signs of damage that may have occurred during transit. Unpack all items and carefully check them against the shipping bill of lading to verify that all components have been received.

Any discrepancies, such as shipping damage or missing items, must be noted on the bill of lading before signing to accept the shipment. Per Interstate Commerce Commission (ICC) requirements, all claims must be made by the consignee. If damage is found, request an immediate inspection by an agent of the shipping carrier and ensure all proper claim forms are executed. Report any damage or shortage claims immediately to the JWT Sales Team.

3.2 Rigging and Handling

DANGER: DO NOT USE A FORK LIFT TO UNLOAD OR MOVE THE RECIRCULATOR PACKAGE. IMPROPER HANDLING CAN CAUSE THE UNIT TO BECOME UNSTABLE AND TIP OVER, RESULTING IN DEATH, SEVERE INJURY, AND CATASTROPHIC EQUIPMENT DAMAGE.

The unit must be moved using a crane with appropriate rigging, operated by a qualified and certified rigger.

1. **Reference Drawings:** Before lifting, consult the certified general arrangement drawing provided by JWT for the exact unit weight and the designated lifting points.
2. **Use Spreader Bars:** Spreader bars of sufficient length and capacity **must** be used across both the length and width of the package. This distributes the lifting forces properly and prevents the lifting slings from applying damaging compressive stress to the vessel, piping, or structural base.
3. **Lifting Points:** Attach shackles only to the designated lifting lugs provided on the unit.
4. **Center of Gravity: WARNING:** The unit may be top-heavy, and its center of gravity is likely not in the geometric center of the package. The lifting operator must use extreme care to check the level and stability of the load. Use balancing chains, cables, or straps to ensure the load does not shift and remains stable and level throughout the lift. Lift the unit only a few inches off the truck bed initially to confirm stability before proceeding with the full lift.

3.3 Foundation and Mounting

The recirculator package must be installed on a solid, level concrete foundation designed to support the full operating weight of the unit (including the weight of the refrigerant charge).

1. **Leveling:** Before anchoring, level the unit on the foundation. Use steel shims or leveling plates under each support foot on the structural base to achieve a level state both lengthwise and crosswise.
2. **Grouting:** After leveling and anchoring, use a high-quality, non-shrinking grout to fill the entire space under all structural base beams. This ensures the unit's weight is continuously and evenly supported by the foundation slab, preventing stress and vibration during operation.

3.4 Piping Installation

WARNING: THIS UNIT IS SHIPPED WITH A DRY NITROGEN HOLDING CHARGE (TYPICALLY 5 PSIG). BEFORE REMOVING ANY PIPE CAPS OR FLANGE BLANKS, SAFELY VENT THIS PRESSURE TO ATMOSPHERE BY SLOWLY OPENING A DRAIN VALVE ON THE PACKAGE. FAILURE TO DO SO CAN RESULT IN THE VIOLENT EJECTION OF CAPS OR PLUGS, CAUSING SERIOUS INJURY.

All piping must be installed in accordance with ASME B31.5, *Refrigeration Piping and Heat Transfer Components*, and all applicable state and local codes.

1. **Pipe Support: CAUTION:** All field-installed piping connecting to the JWT package **must be independently supported**. The vessel nozzles and pump flanges are not designed to support the weight or thermal expansion/contraction stresses of external piping. Imposing external loads on the package connections will lead to mechanical strain, failed welds, and eventual refrigerant leaks.
2. **Piping Slope:** To prevent the trapping of liquid refrigerant or oil, the wet return (inlet) line and the dry suction (outlet) line must be sloped downwards towards the recirculator vessel.
3. **Material Compatibility:** All piping, valves, and fittings used in ammonia service must be constructed of compatible materials, primarily carbon steel (e.g., ASTM A106 Grade B for standard temperatures, ASTM A333 Grade 6 for low temperatures) or stainless steel. **NEVER** use copper, brass, bronze, or other copper alloys in any part of an ammonia refrigeration system, as they will be rapidly corroded by the ammonia.
4. **Leak Testing:** After installation, the new piping must be leak-tested in accordance with

all applicable codes. Isolate the recirculator package from the field piping during high-pressure testing to avoid exceeding the pressure rating of certain components. Safety relief valves must be removed or isolated prior to any pressure testing that exceeds their set pressure and reinstalled afterward.

3.5 Electrical Installation

All electrical installation work must be performed by qualified electricians in accordance with the National Electrical Code (NEC), local codes, and the specific instructions provided in this manual.

1. **Mandatory Reading:** Before beginning any wiring, thoroughly read and understand the procedures outlined in **Appendix C: Proper Installation of Electronic Equipment in an Industrial Environment**. Failure to follow these specific grounding and shielding practices is a primary cause of control system malfunction and VFD failure.
2. **Power Wiring:** Run dedicated power wiring from the motor starter to the pump motors. Wire sizing must meet or exceed NEC requirements for the motor's Full Load Amperage (FLA).
3. **Control Wiring:** Provide a dedicated, isolated 120V AC circuit for the control panel. To minimize instantaneous voltage dips that can disrupt sensitive electronics, size the control power supply wires one gauge larger than required by the NEC for the anticipated amperage draw.
4. **Best Practice - VFD Installation:** The high-frequency electrical noise generated by Variable Frequency Drives (VFDs) can interfere with control systems if not properly managed.
    - **Cabling:** Use shielded, VFD-rated motor cable for the connection between the VFD and the pump motor. If VFD-rated cable is not used, the wiring must be installed in a dedicated, grounded rigid metallic conduit.
    - **Grounding:** The ground conductor for VFD applications must be an insulated, stranded copper wire sized **two sizes larger** than the NEC minimum requirement. This provides a low-impedance path for high-frequency noise to ground, protecting sensitive electronics.
5. **Hazardous Location Classification:** Verify that the machinery room meets NEC hazardous location classification (Class I, Division 2 or Zone 2) unless continuous ventilation or automatic shutdown systems meet IIAR 2 exceptions. Confirm classification before wiring or energizing equipment.

Section 4: Commissioning and Operation

4.1 Pre-Start-Up and Commissioning Checklist

DANGER: DO NOT INTRODUCE AMMONIA INTO THE SYSTEM UNTIL ALL ITEMS ON THIS CHECKLIST HAVE BEEN VERIFIED, COMPLETED, AND SIGNED OFF BY A QUALIFIED TECHNICIAN AND THE FACILITY'S AUTHORIZED REPRESENTATIVE. THIS CHECKLIST IS A CRITICAL SAFETY AND COMPLIANCE STEP ALIGNED WITH ANSI/IIAR 5-2025 PRINCIPLES.

This checklist serves as a formal, auditable record that the system has been installed correctly and is safe to charge and operate. It should be retained as part of the facility's permanent PSM documentation.

Category	Item	Verified (✓)	Comments / Initials
Mechanical	Foundation anchor bolts are properly installed and torqued.		
	All field piping is independently supported; no strain on package connections.		
	All required valves (isolation, control, check) are installed per P&ID.		
	Temporary pump		

	suction strainers are installed for initial start-up.		
	All pressure relief valves are installed, and discharge piping is unobstructed.		
	System has been successfully pressure-tested and evacuated per code.		
Electrical	All power and control wiring is complete and verified against diagrams.		
	All electrical enclosures are closed and secured.		
	Pump motors have been "bumped" to confirm correct rotational direction.		
	VFD parameters (min/max speed, accel/decel, motor data) are correctly		

	programmed.		
	Control power supply is energized and voltage is stable and correct.		
Safety	All required ammonia detectors are installed, calibrated, and functional.		
	Machinery room emergency ventilation system is functional.		
	All required PPE (face shields, goggles, gloves, respirators) is available at the job site.		
	Eyewash/safety shower stations are operational and unobstructed.		
	All personnel are aware of emergency shutdown procedures and muster points.		

4.2 Initial System Charging and Pump Preparation

After the pre-start-up checklist is complete, the system is ready for its initial charge of ammonia.

1. Follow the facility's specific SOP for charging the refrigeration system with liquid ammonia.
2. Introduce liquid into the recirculator vessel until the level is visible in the level column, typically at the normal operating level setpoint.
3. **CRITICAL STEP - PUMP SOAKING:** Once liquid is present in the vessel and the pump suction valves are open, allow the refrigerant pumps to sit idle and "soak" in the liquid ammonia for a minimum of **60 minutes**. This allows the pump's internal components and casing to cool down to operating temperature, preventing thermal shock and ensuring internal clearances are correct. It also allows the liquid to lubricate internal bearings and seals before any rotation occurs.

DANGER: NEVER RUN A REFRIGERANT PUMP DRY. EVEN MOMENTARY OPERATION WITHOUT LIQUID WILL CAUSE IMMEDIATE AND SEVERE DAMAGE TO SEALS, BEARINGS, AND OTHER INTERNAL COMPONENTS, POTENTIALLY LEADING TO A CATASTROPHIC REFRIGERANT LEAK.

4.3 Pump Start-Up Sequence

1. **Verify Valve Positions:** Before starting a pump, confirm the following valve alignment:
   - Pump suction isolation valve: **Fully Open**.
   - Pump discharge isolation valve: **Cracked Open** (approximately one full turn from closed).
   - Motor cooling/recirculation line valve (if equipped): **Fully Open**.
   - Minimum flow bypass line isolation valves: **Fully Open**.
2. **Confirm Rotation:** If not already confirmed during the electrical checkout, jog the pump motor briefly. For open-drive pumps, visually verify the shaft rotates in the direction of the arrow on the pump casing. For hermetic pumps, start the pump for 5 seconds and note the discharge pressure, then stop the pump, reverse two of the three power leads at the starter, and repeat. The correct rotation corresponds to the higher of the two pressure readings. Return wiring to the correct configuration.
3. **Start Pump:** Start the pump via its H-O-A (Hand-Off-Auto) switch in the control panel.

4. **Initial Checks:** Immediately after starting, listen for any unusual noises (e.g., grinding, excessive cavitation) or vibrations. Check the pump discharge pressure gauge to ensure it is stable and showing a pressure rise above suction pressure.
5. **Monitor Amperage:** Using a clamp-on ammeter at the motor starter, verify that the motor's current draw does not exceed the Full Load Amps (FLA) rating on the motor nameplate.
6. **Fully Open Discharge:** Once the pump is running smoothly, **slowly** open the pump discharge valve to the fully open position. Continue to monitor motor amps and system pressures as the valve is opened.

4.4 System Controls and Operation

- **Level Control:** The liquid level in the vessel is managed by an electronic level probe and redundant high-level float switches. The electronic probe provides a continuous signal to the controller, which maintains the operating level by modulating the liquid make-up valve. The float switches provide critical safety cutouts for high- and low-level alarm and shutdown conditions.
- **Liquid Make-Up:** A modulating valve, controlled by a proportional signal from the main controller, automatically feeds liquid refrigerant into the vessel to replace what has been evaporated in the system evaporators.
- **Minimum Flow Bypass:** A bypass line is installed from the pump discharge back to the vessel. This line includes a hand expansion valve or regulator. This bypass must be set to ensure that even when all system loads are satisfied, the pump maintains its minimum required flow rate. Operating a centrifugal pump below its minimum flow can cause overheating, vibration, and cavitation, leading to damage.

4.5 Best Practices: Optimizing Performance

The true value of a modern recirculator package lies in its ability to operate efficiently across a wide range of conditions. The operational philosophy should shift from simple on/off control to dynamic, data-driven optimization.

- **VFD Operation for Energy Efficiency:** The most significant energy savings are achieved by utilizing the Variable Frequency Drive (VFD) correctly. The traditional method of running the pump at full speed and bypassing excess flow through the minimum flow loop is mechanically simple but extremely wasteful of energy. The JWT operational best practice is to use the VFD to match pump speed directly to the system's cooling demand. This is typically achieved by using a pressure transducer on the pump discharge header to provide a feedback signal to a PID control loop. The VFD will automatically adjust the pump's speed to maintain a constant differential pressure between the liquid supply and suction headers, ensuring the evaporators always receive adequate flow while using the absolute minimum amount of pump energy. The minimum flow bypass loop should still be set for pump protection but should see little to no flow during normal VFD-controlled operation.

- **Advanced Sensor Integration & Data Utilization:** Modern industrial refrigeration is increasingly data-driven. The advanced sensors and controls on the JWT package are not just for alarms and shutdowns; they are tools for performance tuning and predictive maintenance. Operators should be trained to monitor key data points from the VFD and control system, such as motor amperage, torque, speed, and energy consumption (kWh). Trending this data over time can reveal inefficiencies, diagnose developing problems before they cause a failure, and provide the basis for continuous system optimization. For example, a gradual increase in motor amps for a given pump speed could indicate bearing wear or fouling in the system, allowing for proactive maintenance scheduling.

Section 5: Maintenance and Troubleshooting

5.1 Routine Maintenance Schedule

A consistent preventative maintenance program is critical for ensuring the safety, reliability, and longevity of the JWT Recirculator Package. All maintenance activities must be performed by trained and qualified technicians. The following schedule is a minimum guideline based on the requirements of ANSI/IIAR 6-2025; the actual frequency may need to be adjusted based on operating conditions, run hours, and facility-specific requirements. All maintenance activities must be documented as part of the facility's PSM Mechanical Integrity program.

Frequency	Component	Task Description	Reference
Daily	Pumps / Motors	Visually inspect for leaks. Listen for abnormal noise or vibration.	5.4
	Vessel	Check liquid level in the sight glass; compare with controller reading.	4.4
Monthly	Control Panel	Verify indicator lights are functional. Check for any active alarms.	5.4
	Level Controls	Manually lift float switch linkages to test high-level alarm and cutout functions.	4.4

Quarterly	Pump/Motor	Check pump/motor alignment (for open-drive models).	Mfr. Manual
	Oil Pot	Drain oil from the system as needed.	5.2.1
Annually	All Stop Valves	Exercise all service valves through their full range of motion to ensure operability.	-
	Sensors/Probes	Verify calibration of pressure transducers and electronic level probes.	Mfr. Manual
	Safety Systems	Test and document the functionality of all safety cutouts (low level, high level).	4.4
Per Code	Relief Valves	Replace pressure relief valves in accordance with IIAR and jurisdictional requirements (typically every 5 years).	-

5.2 Critical Maintenance Procedures

5.2.1 Oil Removal Procedure

DANGER: DRAINING OIL FROM AN AMMONIA SYSTEM IS A HIGH-RISK TASK. AN UNCONTROLLED RELEASE OF HIGH-PRESSURE LIQUID AMMONIA AND OIL CAN OCCUR. WEAR A FULL FACE SHIELD, CHEMICAL GOGGLES, AND AMMONIA-RATED GLOVES. NEVER WORK ALONE.

The oil pot is designed to collect lubricating oil that migrates to the low-pressure side of the system. The frequency of draining depends on the efficiency of the high-side oil separators and must be determined by the plant operator.

1. **Isolate the Oil Pot:** Close the oil/liquid inlet valve that feeds the oil pot from the recirculator vessel.
2. **Allow Warming:** Slightly open the vent valve during the 30-60 minute warm-up to safely release vapor into the vent header, then close the vent before draining. **WARNING:** Monitor the pressure gauge on the pot at all times. **NEVER** apply external heat (e.g., a torch or hot water) to the oil pot, as this can cause a dangerous pressure increase.
3. **Prepare for Draining:** Connect a high-pressure ammonia-rated hose to the spring-return drain valve. Route the other end of the hose to a suitable, vented container approved for ammonia-contaminated oil.
4. **Drain Oil:** First, fully open the oil drain isolation valve. Then, **slowly and carefully** actuate the spring-return drain valve to meter the oil out of the pot. **NEVER** use the isolation valve to throttle the flow.
5. **Secure the System:** Once all oil is removed, release the spring-return valve to let it close. **Immediately close the isolation valve first.** This sequence allows any refrigerant trapped in the drain piping to vent back into the pot, preventing it from being trapped between the two valves.
6. **Return to Service:** Open the vent valve to equalize pressure with the main vessel. Finally, slowly open the oil/liquid inlet valve to return the oil pot to normal operation.

5.2.2 Pump Suction Strainer Service

Temporary suction strainers are installed during commissioning to capture construction debris. They must be removed after the first 6-8 hours of operation to prevent flow restriction and potential pump cavitation.

1. **Isolate the Pump:** Start the standby pump and verify it is operating correctly. Stop the pump to be serviced and close its suction and discharge isolation valves. Follow all LOTO procedures.
2. **Purge Ammonia: WARNING:** The isolated pump section contains trapped liquid ammonia under pressure. Safely vent this pressure and purge the liquid to an appropriate recovery system or flare using the pump's drain/vent valve.
3. **Remove Strainer:** Once the section is verified to be at 0 psig and free of liquid, unbolt the suction spool piece flanges. Remove the temporary conical strainer.
4. **Install Spacer:** Clean the flange faces. Install a correctly sized spacer plate with new gaskets on both sides.
5. **Reassemble:** Re-bolt the spool piece, tightening the bolts in a star pattern to the specified torque.
6. **Return to Service:** After reassembly, the pump can be leak-tested and returned to standby service.

5.3 Best Practices: Predictive Maintenance

Move beyond scheduled, time-based maintenance by leveraging the data provided by the JWT control system. Condition-based monitoring allows for the early detection of developing faults, reducing unplanned downtime and preventing catastrophic failures. Operators and technicians should be trained to recognize these data-driven cues.

Observable Data from Control System / VFD	Potential Cause	Recommended Proactive Action
Pump motor amps trending slowly upwards over weeks at a constant speed.	Progressive bearing wear; increased system friction; motor winding degradation.	Schedule vibration analysis on the pump and motor. Inspect system for any new sources of pressure drop.

VFD frequently trips on "Over-Torque" during start-up.	Pump is attempting to start against a closed discharge valve; liquid in pump is partially frozen (slush); oil is too viscous at low temp.	Verify valve positions and start-up sequence. Check for issues with oil pot heater (if equipped).
Pump speed (RPM) is hunting or oscillating excessively to maintain pressure.	System load is highly unstable; PID loop is poorly tuned; air or non-condensables in the system.	Investigate source of load swings. Consult JWT for PID tuning assistance. Check for non-condensables.
Pump suction pressure fluctuates rapidly.	Cavitation due to insufficient NPSH; vortexing in the vessel at low liquid levels.	Verify vessel liquid level is above the minimum required for pump operation. Check for restrictions in the suction line.

5.4 Troubleshooting Guide

This guide provides a logical starting point for diagnosing common operational issues. Always follow proper safety and LOTO procedures before performing any corrective actions.

Symptom	Potential Causes	Corrective Actions
Pump is Noisy / Vibrating	1. Cavitation (insufficient NPSH). 2. Mechanical misalignment (open-drive). 3. Worn bearings. 4. Trapped vapor in pump volute.	1. Verify liquid level in vessel is sufficient. Check for restrictions in suction line (clogged strainer). 2. Perform a precision alignment. 3. Schedule pump for overhaul. 4. Vent

		the pump casing.
Low Discharge Pressure	1. Pump running in reverse. 2. Worn pump impeller. 3. System demand exceeds pump capacity. 4. Large system leak.	1. Correct motor wiring for proper rotation. 2. Overhaul or replace pump. 3. Verify system design. Check for fully open hand expansion valves. 4. Isolate and repair leak.
High Liquid Level Alarm / Cutout	1. Liquid make-up valve leaking or stuck open. 2. Large, rapid return of liquid from hot gas defrost. 3. Faulty level probe or float switch.	1. Isolate, purge, and repair/replace the make-up valve. 2. Adjust defrost schedules to stagger large loads. 3. Test, recalibrate, or replace level controls.
VFD Fault Code	1. Refer to VFD manufacturer's manual for specific code. 2. Common faults: Overcurrent, Overvoltage, Undervoltage.	1. Follow VFD manual's troubleshooting steps. 2. Check for motor short/ground. Check incoming power quality. Verify VFD parameters.

Section 6: Technical Appendices

6.1 Appendix A: Piping and Instrumentation Diagrams (P&IDs)

For the P&ID, please contact JWT at ewright@jwtank.com

6.2 Appendix B: Electrical and Control Diagrams

For the Wiring Diagram, please contact JWT at ewright@jwtank.com

6.3 Appendix C: Proper Installation of Electronic Equipment in an Industrial Environment

The industrial environment is electrically noisy due to large motors, VFDs, and other equipment. Electronic controls are susceptible to this noise (EMI/RFI) and to grounding issues, which can cause erratic behavior, false readings, and component failure. Adherence to the following practices is mandatory for a reliable installation.

- **Voltage Source:** The control panel must be powered from a dedicated, isolated control power transformer. Do not power the panel from a general-purpose lighting or utility panel, as this will expose it to significant electrical noise from other plant equipment.
- **Wire Sizing:** Size control power conductors one gauge larger than the NEC minimum for the expected load. This reduces voltage sag during the inrush of contactor coils and solenoids, which can cause microprocessor resets.
- **Critical Grounding Practices:** Improper grounding is the single most common cause of electronic control problems.
  - **Single-Point Ground:** The control system requires a single, continuous, insulated copper ground wire from the panel's ground bus back to the ground point at the secondary of the control power transformer.
  - **No Daisy-Chaining:** Never "daisy-chain" ground wires from one control panel to another. Each panel must have its own dedicated ground conductor back to the source.
  - **Conduit is Not a Ground:** Do not rely on metallic conduit as the primary ground conductor for electronic controls. While the conduit must be grounded for safety, a dedicated insulated copper wire is required for noise-free operation.
- **VFD-Specific Considerations:** VFDs are a major source of EMI.

    - **Wiring:** Use shielded, VFD-rated cable or run motor leads in grounded rigid metallic conduit. Maintain maximum separation between power and control wiring.
    - **Grounding:** The VFD motor ground must be an insulated copper wire sized two gauges larger than the NEC minimum. This provides the lowest impedance path for high-frequency noise.
- **Wiring Separation:** Never run low-voltage DC analog or communication wires in the same conduit as AC power wires (including 120V AC). Maintain a minimum separation of 12 inches between parallel runs of power and control conduits.

6.4 Appendix D: Recommended Spare Parts List

It is recommended that the facility maintain a stock of critical spare parts to minimize downtime.

- **Critical (Keep On-Hand):**
    - Complete spare refrigerant pump.
    - Pump mechanical seal kit.
    - Pump bearing kit.
    - Fuses for control panel.
    - Level probe and float switch assembly.
- **Recommended:**
    - Pump/motor coupling element.
    - Set of gaskets for all flanged connections.
    - Liquid make-up valve repair kit.

6.5 Appendix F: Decommissioning

At the end of the equipment's service life, the ammonia must be safely removed and the equipment decommissioned. All procedures must be performed in accordance with **ANSI/IIAR 8-2020, *Decommissioning of Closed-Circuit Ammonia Refrigeration Systems***, and all applicable federal, state, and local regulations. This typically involves transferring the ammonia charge to a suitable storage or transport vessel using a recovery compressor. Only qualified technicians experienced in ammonia handling should perform decommissioning procedures.